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Next item →

1. In the context of Keras, what is the primary purpose of a transpose convolution layer in a model? 1 / 1 point

- ☐ To perform max pooling operations.
- ☐ To reduce the computational complexity of the model.
- ☒ To increase the spatial resolution of the input image.
- ☐ To combine multiple input feature maps into a single output feature map.

✔ Correct
Correct! The transpose convolution layer, also known as deconvolution, is primarily used to increase the spatial resolution of the input image.

2. Which of the following statements describe the interdependent relationship between the generator and discriminator in a GAN? 1 / 1 point

- ☒ The generator and discriminator directly compete with each other.

✔ Correct
Correct! The generator and discriminator are in a direct competition, where the generator tries to create realistic samples and the discriminator tries to distinguish them from real samples.

- ☐ The generator and discriminator are trained in isolation.
- ☐ Both the generator and discriminator aim to minimize the same loss function.
- ☐ The success of the generator does not affect the discriminator's training.
- ☒ The discriminator provides feedback to help the generator improve.

✔ Correct
Correct! The discriminator's feedback on the generated samples helps the generator improve its output.

3. During the training process of a GAN, what does the discriminator aim to achieve? 1 / 1 point

- ☐ To produce data samples indistinguishable from real data.
- ☐ To generate random noise for the generator.
- ☒ To minimize the loss for real and fake data samples.
- ☐ To train the generator to produce better data samples.

✔ Correct
Correct! The discriminator aims to accurately classify real data as real and fake data as fake, thereby minimizing the loss for both types of samples.

4. Which of the following are characteristics of the MNIST dataset? 1 / 1 point

- ☐ Images are in RGB format
- ☒ Contains images of handwritten digits

✔ Correct
Correct! The MNIST dataset contains images of handwritten digits.

- ☐ Primarily used for natural language processing tasks
- ☒ Has 10 classes

✔ Correct
Correct! There are 10 classes in the MNIST dataset, representing digits 0-9.

- ☒ Each image is of size 28x28 pixels

✔ Correct
Correct! Each image in the MNIST dataset is of size 28x28 pixels.

5. In the context of GANs, what is the primary purpose of the discriminator? 1 / 1 point

- ☐ To reduce the dimensionality of the dataset
- ☐ To optimize the generator's loss function
- ☒ To differentiate between real and generated data
- ☐ To generate high-quality images

✔ Correct
Correct! The primary purpose of the discriminator is to differentiate between real and generated data.